

Aesthetics, function and CAD/CAM: a complete case

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Digital processes and software used

- Nikon 610 camera and R1C1 flash
- Omnicam® Optical Camera (Sirona)
- Nemotec® software for 2D and 3D rendering of the anteriors.
- Modjaw® system for 4D analysis of mandibular movements

CAD software

- Cerec Inlab®
- Exocad®



Preoperative view.



Work in progress.



Postoperative view.

1

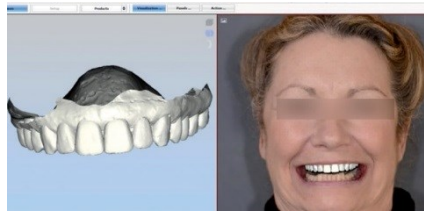


Preoperative digital study

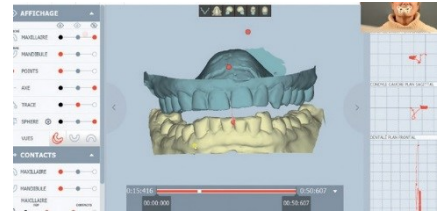
Left lateral preoperative view.



2D analysis: the initial photographic reports allow 2-dimensional project simulation.



3D analysis: superimposing the STL model makes it possible to concretize the first approach from an aesthetic point of view.



4D analysis: integration of mandibular movements make it possible to take functional constraints into consideration.

2



Project validation, anterior guide and aesthetics

Left lateral view with complete mock-up.



Complete occlusion transformation in a single session.
In this case, the mock-up was created with a precise silicone core duplicating the printed model of the prosthetic project.
The mock-up was kept for 3 weeks.



The anterior guide is made by maxillary palatal facets and mandibular vestibular facets. Here we see the fitting of a palatal facet on a printed model.



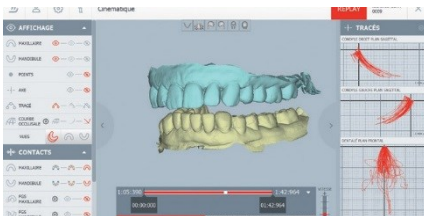
The maxillary vestibular facets are created in a completely digital manner and united by gluing.

3



Functional reevaluation and creation of the posteriors

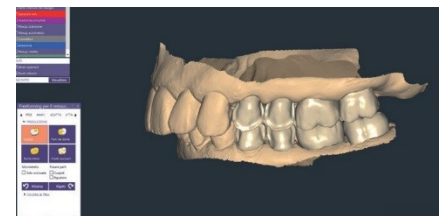
Side view with the usual restorations.



4D CAD/CAM functional reevaluation using the Modjaw technique. Functional path recording with the existing anterior guide makes it possible to refine the posterior anatomies.



Digital impression of posterior teeth with the Omnicam camera.



Exocad® can re-animate the functional paths recorded by Modjaw®, thus improving anatomies during the design and of course the conception of the posterior prostheses.

1



Preoperative study Digital creation of the prosthetic project

2D analysis



The patient has a desire for aesthetic improvement and complains of functional discomfort. The initial photo report gives the patient a detailed view of his/her own mouth, communication is facilitated – the analysis begins.

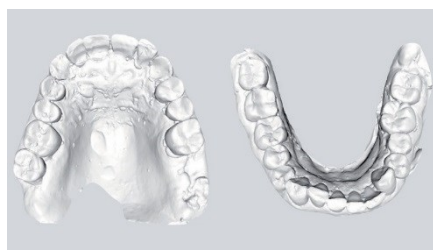


The set of DSD analysis technical photos, along with the video, provide all the elements needed for an initial approach.

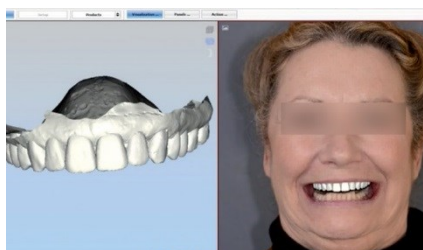


Creating the project in 2D, here using Nemotec®, makes it possible to project the result and to plan the treatment sessions.

3D analysis



The initial impressions are digitized in STL format, either directly from an optical camera or after a scan of the initial plaster model.

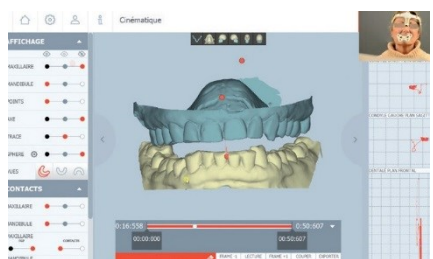


This 3D file is superimposed on the photographs, giving rise to a virtual model of the aesthetic project.

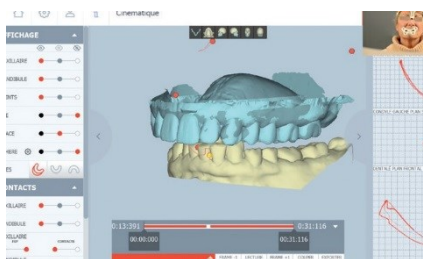


Video realization of the 3D project before mock-up/after mock-up. Actual aesthetic evaluation of the 3D result and validation by the patient.

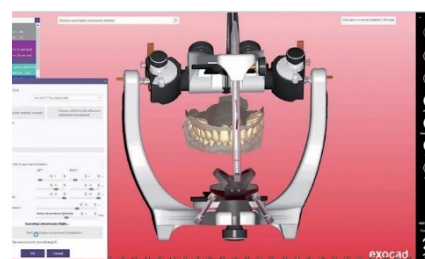
4D analysis



Modjaw® analysis of the patient's function: functional path recording, search for centred relation, analysis of posterior determinants, any exercises asked of the patient are recorded in 4D.



Aesthetic project reanimation in Modjaw®: here we can directly see the impact of the aesthetic proposition on the function envelope. At this point, we can fully associate functional considerations with aesthetics.



Choice of the new DVO, data transfer by digital facebow and programming of the digital articulator: this is the virtual creation of the complete prosthetic project.

2

Validation and creation of the anteriors



Creation of the complete mock-up



This first prosthetic project is printed. Here, 3D printing is performed with the Formlabs® printer.



Precise silicon condensation cores, here Président®, are created on the models.



At the end of the first work session, all the intermaxillary ratios are transformed, no preparation has been made. The project is tested in a real-life situation from a functional and aesthetic point of view.

Creating the anterior guide



Preparation of maxillary palatal facets and mandibular vestibular facets. These restorations will ensure calibration, centring and guiding. Posterior setting is ensured by the mock-up resin.



Digital impression of preparations with the Omnicam® camera. Modelling is carried out with Cerec Inlab® software after superimposing the impression onto the initial project.



Union by facet gluing. The palatal facets are made from Vita Enamic® and the vestibular facets from Vitabloc Trilux A1®.

Anterior maxillary aesthetic realization



Preparation of the maxillary anterior teeth, digital impression, modelling with Exocad® after superimposition onto the initial project, machining of VitaTrilux® blocks and repositioning on Formlabs® printed model.



Nictone® dam bonding by flying clamp technique with Optibond FL® Kerr adhesive and Variolink LC® Ivoclar Vivadent resin.



Situation of the patient upon removal of the operative field.

3



Functional reevaluation and posterior staging

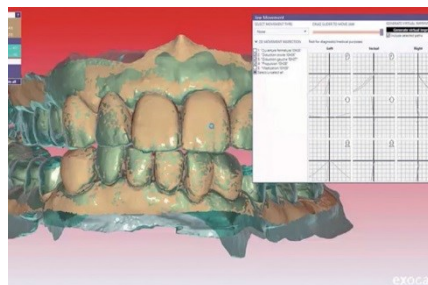
Functional reevaluation: After balancing the new anterior guidance, in this new DVO, the patient was able to function and integrate her new function envelope.



4D analysis in Modjaw® is performed once more to record the new masticatory paths as and all other occlusal parameters.



Targeted exercises make it possible to choose the exports to replicate in CAD software, masticatory cycles, swallowing and excursions.



The paths recorded under Modjaw® are accurately replicated under Exocad®. The initial virtual drawing is improved by collecting this basic information.

Removal of old prostheses and impressions



Removal of old crowns and digital impression of preparations. Note that access to the boundaries of peripheral crowns is more complicated by optical camera than by conventional impression... this is another subject.

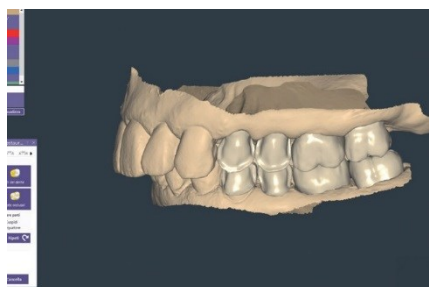


Impression with the Omnicam® camera. At this stage, no. 17 is being osseointegrated, no. 27 being endodontically treated, while the persistent no. 48 on the arch facilitates arch superimposition.



CAD of the posterior crowns under Exocad® by integrating the patient's true functional paths. The articulator is no longer necessary, the digital avatar is used.

Creation of the posterior crowns



Monobloc crowns are drawn on the posteriors, the premolars are CIVs, allowing stratification for aesthetic purposes.



The crowns are made of Noritake Katana® Zirconia material and are repositioned on a model printed by the Formlabs® printer for manual finishing.



Postoperative maxillary and occlusal contacts view immediately after sealing and before balancing.

4

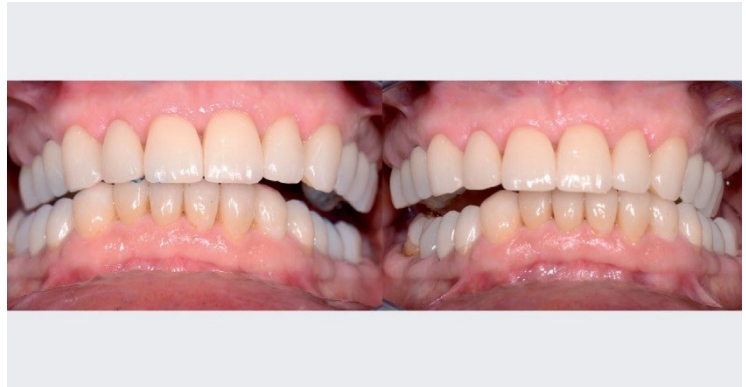


Postoperative situation: assessment and critiques

Postoperative situation at one week. The initial calibration was respected thanks to the digital sequence. Note the slight excess of brightness of the posterior zirconias compared with the restored glued glass-ceramics.



Happy smile of the patient who finds the ease of an aesthetic and functional mouth.



The Planas functional mastichatory angles are identical, the lateral translations of the guide cusps are tangential in excursion and the non-working opening is symmetrical.

Thanks to digital processes, prosthetic therapies are constantly improving.

These improvements relate to diagnostic quality, control of therapeutic processes and, above all, they save time, both for the patient, and for the therapist and laboratory.

Work sequences follow a literature-based logic. The treatment plan presented here is obviously based on diagnosis [1], which adopts an aesthetic starting point [2]. The relationship between aesthetics and function is decided by 4D technology [3], and if the vertical dimension is adapted to the prosthetic project [4, 5], the new intermaxillary relationship uses conventional techniques [6] and enjoys digital recording accuracy [7], all with the aim of improving mastichatory function [8]. The goal is a minimally invasive procedure [9] allowing prosthetic-guided preparations and ensuring the best quality [10] of dental tissues to support prostheses when stuck.

The digital stream presented here was performed at the end of 2017. Since then, new breakthroughs have been made. The constant updating of our knowledge is a chance for our profession, we will doubtless need to support our entire professional sector in this digital shift.

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